HAIG-SIMONS INCOME*

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Abstract

TBD

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1 Introduction

The year 2021 yielded \$16.2 trillion in aggregate capital gains on assets held by households in the United States — 94% of net national income, and an amount larger than wages, dividends, or interest income. While 2021 returns were unusually large, economically signifcant gains have become increasingly common: over the past two decades, real asset appreciation averaged 20% of national income. The rise of capital gains presents challenges for both macroeconomic accounting, where capital gains are not included in national income, and taxation, where capital gains are uniquely subject to a realization rule (they are taxed only when assets are sold). Capital gains are particularly prevelant amoung the wealthy, and have been the subject of numerous recent policy proposals.[1]

Recent scholarship has explained the rise in asset values by secular changes in earnings and profits (**?**), driven by monopoly rents (**??**), retained earnings, intangible investment (**?**), a shift in income to the corporate sector (**?**), and a rise in free-cash flow relative to earnings (**?**). The rise in capital gains have in turn been linked to widespread macroeconomic consequences: higher spending and local employment in response to stock market returns (**?**), greater household wealth (**?**), and elevated leverage prior to the Great Recession (**?**). Less is known, however, on *who* receives the gains— which shapes not only the strength of such aggregate effects but also directly affects measures of the income and wealth distribution.

The primary goal of this paper is to estimate the distribution of capital gains by asset class at the individual level, and quantify their relationship with income, wealth, realized capital gains, and demographic groups. We then apply our new estimates to three key policy-relevant questions: how capital gains affect measures of income inequality, the effective tax rate on capital gains and comprehensive income, and the impact of gains on wealth inequality.

We do so using internal tax data from 2002-2021. The advantage of tax data is its comprehensiveness, spanning the entire income and wealth distribution. The disadvantage is that only realized sales of assets directly show up on tax forms, which are only a small fraction of total appreciation (the sum of unrealized and realized). Prior research has found only about 1/3rd of the stock of capital gains have been realized.[2]

To estimate total (unrealized and realized) gains at the individual level, we use a three step procedure: (i) link individuals to their specific portfolio holdings (ii) capitalize income (using heterogenoeus capitalization factors when available) to estimate wealth (iii) estimate capital appreciation by multiplying wealth by asset class specific returns (using heterogeneous returns when available). In doing so, we exploit never before used tax forms, new data sources, and im-

[1]These include recent Republican proposals to exempt owner-occupied housing from gains and index gains to inflation, as well as Democratic proposals to tax unrealized capital gains and eliminated the step-up basis at death.

[2]See **?** for Norway, **?** and **?** for the United States.

proved estimation techniques to provide the most accurate available measures of wealth and capital gains at the individual level. We highlight three innovations here: (i) we use techniques from the network economic literature to better track partnership wealth through network chains of ownership, which allows us to trace 90% of partnership assets to their ultimate owners (ii) we use never before used tax forms to link 100% of rental real estate assets to their ultimate owner, which allows for a more accurate estimate of real estate wealth and capital gains (iii) we use a new data source on private business sales to better estimate business valuations and capital gains.

We document several new empirical facts. First, comparing taxable realized gains and aggregate revaluations, we find less than 20% of total gains were subject to taxation. This low rate is driven primarily by individuals delaying realization, as well as a growing share of assets held in tax exempt accounts. The correlation of capital gains and tax realizations is only XXX. Demographically, capital gains concentrated amongst the elderly, with gains rising linearly in age. As the population ages across our sample period, so does the concentration of capital gains in the aged. Amongst individual filers, men have substantially higher gains than women. Across geography, [states that have high KGs]. Their correlation with labor income, capital income, wealth, etc.

The distribution of capital gains is right-skewed, with higher concentration than any other form of income: the top 1% of the distribution received 45% of gains over the sample period. They are concentrated amongst the upper tails of the income and wealth distribution: the top 1% (10%) received 45.3% (75.7%) of total revaluations over our sample period. The wealthiest 1% received 90% of capital gains, while holding only 60% of wealth.

We estimate the distribution of a comprehensive measure of Haig-Simons income that is inclusive of capital gains. Because capital gains are highly concentrated, they increase measured income inequality. The top 1% (10%) of individuals received 21.0% (47.9%) of Haig-Simons income; their share without capital gains is 18% (45%). Capital gains on public and private equity are the main driver of the increased concentration, while housing and pension capital gains lead to a more equal distribution of income.

We estimate the effective tax rate on aggregate capital gains and Haig-Simons ncome. Althought the average marginal statutory tax rate on realized gains was 18%, due to low realization rates the the average effective nominal tax rate on total gains was just 3% and 5% for real gains. Low tax rates on capital gains in turn lower the tax rate on Haig-Simons income. Surprisingly, high income and wealth groups realize gains at a someone higher rate, and this combined with higher statutory rates lead to higher tax rates on total capital gains. However, since high income and wealth groups have a larger share of gains, their Haig-Simons tax rate is pulled down. As a result, the tax-system is substantially less progressive when measured by Haig-Simons income, with tax rates roughly flat across the income distribution.

Our results on the effects of income concentration and tax progressivity are robust to a variety of alternative assumptions about the allocation of labor and capital income, as well as the incidence of taxes and government spending. While the *level* of income concentration depends on whether we distribute non-capital gain income according to **?** or **?**, capital gains always substantially raises measured income inequality. Our results are also robust to a variety of alternative measurement assumptions for capitalization rates, returns, and unmeasured income.

Finally, we show that the secular increases in stock prices, business values, and housing prices substantially increased the welfare of asset holders over our time period. A long-standing criticism of Haig-Simons income is that when asset price changes are caused by declining interest rates and/or changes in discount rates, individual welfare cannot be computed from the change in asset prices, since in addition to the endowment effect of higher asset values there is an offsetting effect of future lower returns. In a series of robustness exercises, we show, consistent with prior research, the majority of long-run capital gains over the sample period were caused by changes in dividend flows rather than interest rate changes and/or discount rate changes. In addition, we construct capital gains series purged of interest rate and discount rate variation, and show that the resulting distributional series are largely unchanged.

2 The Demographics of Capital Gains

Capital gains are concentrated among older individuals, reflecting the cumulative nature of asset accumulation over the life cycle. Figure 1 displays average capital gains by age, which rise steadily from young adulthood through retirement. The age-gains profile is approximately linear through working ages before leveling off after age 70. Notably, capital gains do not decline substantially among the oldest age groups, consistent with the well-documented failure of wealth to decumulate in retirement as predicted by standard life-cycle models (**?**). Bequest motives, precautionary saving against health and longevity risk, and the concentration of wealth among households with low marginal propensities to consume all contribute to the persistence of wealth—and hence capital gains—into advanced ages.

graphs/kg_age_profile.png

Figure 1: Average Capital Gains by Age

Notes: Figure displays average annual capital gains by 5-year age bracket, pooled over 2002–2021. Capital gains include unrealized and realized appreciation on equities, housing, pensions, and private businesses. Shaded area indicates 95% confidence interval. Sample restricted to tax units with positive wealth.

Table 1 summarizes the distribution of capital gains across three broad age groups. Given the strong age-relationship documented above, capital gains are concentrated amongst retirees: individuals aged 65 and over received 36% of total capital gains while comprising only 18% of the adult population. Individuals aged 20-44 comprise 48% of the population, but received only 20% of the gains. The average age of individuals in the top 1% of capital gains recipients is [XX] years, compared to [YY] years for the median recipient. Among the top 0.1%, the average age rises further to [XX] years. The concentration of capital gains among the elderly is driven primarily by stock market wealth. Individuals aged 65 and over received 45% of stock gains—far exceeding their [XX]% population share. By contrast, the elderly are only modestly overrepresented amongst real estate and private business gains.

Table 1: Distribution of Capital Gains by Age Group

Age Group	Mean KG ($)	Share of Gains	Share of Gains by Asset: Stock	Business	Housing	Share of Pop.
20–44	[XX]	[XX]%	[XX]%	[XX]%	[XX]%	[XX]%
45–64	[XX]	[XX]%	[XX]%	[XX]%	[XX]%	[XX]%
65+	[XX]	[XX]%	[XX]%	[XX]%	[XX]%	[XX]%
All	[XX]	100%	100%	100%	100%	100%

Notes: Distribution of capital gains by age group, 2002–2021. Stock includes public equity and mutual funds. Business includes partnerships, S-corps, and private C-corps. Housing includes owner-occupied and rental real estate.

Table 2 reports the average age and capital gains for individuals at different points in the capital gains distribution. Average age rises monotonically with capital gains: the top 10% of capital gains recipients are [XX] years old on average, compared to [XX] years for the top 0.01%. This pattern reflects both life-cycle wealth accumulation and cohort effects in stock market participation.

Table 2: Characteristics of Top Capital Gains Recipients

Percentile Group	Average Age	Average Capital Gains ($)
Top 10%	[XX]	[XX]
Top 1%	[XX]	[XX]
Top 0.1%	[XX]	[XX]
Top 0.01%	[XX]	[XX]

Notes: Table reports average age and capital gains for individuals in each percentile group, pooled over 2002–2021. Percentiles based on annual capital gains.

The aging of the U.S. population has contributed to increased concentration of capital gains among the elderly over our sample period. Between 2002 and 2021, the share of capital gains received by individuals aged 65 and over rose from 23% to 37%, tracking the growth in this age group's share of total wealth. This demographic shift has implications for both the aggregate consumption response to asset price changes—as older households have lower marginal propensities to consume—and the design of capital gains taxation, particularly provisions such as the step-up in basis at death.

Table 3 shows the breakdown of gains for married versus unmarried filers, and amongst the unmarried, female versus male. Average capital gains are substantially higher for married filers than for unmarried filers, reflecting large differences in income and wealth. Among unmarried filers, men have moderately higher average capital gains than women, again reflecting differences in wealth: [XX] compared to [YY]. The largest differences in gains between men and women concern private businesses, with average male capital gains double that of females. This pattern reflects the well-documented gender gap in entrepreneurship: men are roughly twice as likely as women to start and own businesses (**?**).

Table 3: Income, Wealth, and Capital Gains by Filing Status and Gender

Group	Income ($)	Wealth ($)	Capital Gains ($)
Married	[XX]	[XX]	[XX]
Unmarried Male	[XX]	[XX]	[XX]
Unmarried Female	[XX]	[XX]	[XX]

Notes: Table reports average income, wealth, and capital gains by filing status and gender, pooled over 2002–2021. Married includes all joint filers. Unmarried includes single, head of household, and married filing separately.

2.1 Geography

Capital gains are geographically concentrated in states with concentrations of financial assets appreciating housing markets. Figure 2 displays capital gains per capita by state, averaged over 2002–2021. The geographic distribution is highly uneven: residents of the District of Columbia received an average of $[19,500] in annual capital gains – more than double the national average – while residents of West Virginia received just $3,500. Capital gains are substantially more geographically concentrated than other forms of income: the Gini coefficient for capital gains across states zones is XXX, compared to YYY for other forms of income. Capital gains per person (and taxable capital gains per person) are moderately correlated with state level capital gain tax rates; this survives controls for income.

graphs/kg_map_total.png

Figure 2: Capital Gains Per Capita by State

Notes: Figure displays average annual capital gains per capita by state, pooled over 2002–2021. Capital gains include unrealized and realized appreciation on equities, housing, pensions, and private businesses. States grouped into quintiles.

Table XXX summarizes the correlation of commuting-zon level characteristics and capital gains per person. Capital gains are substantially larger in urban areas than rural areas, “super star cities”, areas with a high share of elderly population, low social mobility,

Table 4: Top and Bottom States for Capital Gains Per Capita

Top 5 States			Bottom 5 States		
Rank	State	KG/Capita ($)	Rank	State	KG/Capita ($)
1	Virgin Islands	[XX,XXX]	1	West Virginia	[X,XXX]
2	District of Columbia	[XX,XXX]	2	Mississippi	[X,XXX]
3	California	[XX,XXX]	3	Indiana	[X,XXX]
4	New York	[XX,XXX]	4	Kentucky	[X,XXX]
5	Florida	[XX,XXX]	5	Alabama	[X,XXX]
National Average:					[XX,XXX]

Notes: Table reports states and territories with highest and lowest average capital gains per capita, pooled over 2002–2021.

Table 5: Top 5 States for Capital Gains Per Capita by Asset Class

Rank	Overall	Equity	Real Estate	Business
1	Virgin Islands	DC	DC	Wyoming
2	DC	Connecticut	California	American Samoa
3	California	Wyoming	Hawaii	DC
4	New York	New York	Washington	Connecticut
5	Florida	Massachusetts	Florida	New York

Notes: Table reports states and territories with highest capital gains per capita by asset class, pooled over 2002–2021. Equity includes public stocks and mutual funds. Real estate includes owner-occupied and rental property. Business includes partnerships, S-corps, and private C-corps.

The geographic concentration of capital gains has important implications for state tax revenues. Because most states tax capital gains as ordinary income, states with large concentrations of capital gains experience pronounced revenue volatility tied to asset market performance. California's Legislative Analyst's Office estimates that capital gains are nearly three times as volatile as the state's overall personal income tax base, and account for roughly 40% of total income tax volatility (**?**). More broadly, personal income tax revenue volatility has increased across the majority of states in recent years, driven in significant part by greater reliance on increasingly volatile capital gains realizations (**?**). California and New York—the two states with the largest concentrations of capital gains—accounted for the majority of the national decline in state income tax revenues in fiscal year 2023 (**?**).

Beyond revenue volatility, the geographic concentration of capital gains implies that standard measures of regional income divergence understate the true extent of spatial inequality. Most studies of geographic income convergence—including the influential work of **?**—measure income using wages or pre-tax factor income that excludes capital gains. Because capital gains are concentrated in already-prosperous states such as California, New York, and Connecticut, incorporating gains into a comprehensive income measure would substantially widen measured geographic divergence. **?** documents the growing concentration of high-skill, high-wage employment in a small number of "superstar" cities; our data suggest that these same cities are also the primary recipients of capital gains, compounding the divergence in economic fortunes across regions.

The geographic concentration of real estate capital gains has particularly important implications for housing affordability and spatial misallocation. In supply-constrained housing markets, asset appreciation accrues primarily to incumbent homeowners, widening the divide between owners and renters and reinforcing barriers to geographic mobility. **?** estimate that land-use restrictions in high-productivity cities such as New York, San Francisco, and San Jose lowered aggregate U.S. GDP growth by more than a third between 1964 and 2009 by preventing workers from relocating to their most productive locations. Housing capital gains in these markets function as a transfer from future buyers to current owners, raising the entry cost for workers seeking access to high-wage labor markets. Recent work by **?**, however, challenges the supply-constraint explanation, finding that income growth—not housing supply elasticity—is the primary predictor of cross-city house price growth over 1980–2020. If income growth rather than regulation drives geographic price divergence, the omission of capital gains from standard income measures is all the more consequential: capital gains represent a large and geographically concentrated component of comprehensive income that is absent from the income measures used in housing demand models. The geographic pattern in our data—with California, Hawaii, Washington, and the District of Columbia leading in real estate gains per capita—is consistent with both supply-constraint and income-driven explanations of housing price divergence, and underscores the need for further research disentangling these channels.

The spatial distribution of capital gains also intersects with intergenerational mobility and opportunity. **?** document substantial geographic variation in rates of upward mobility across commuting zones, driven in part by local conditions such as school quality, social capital, and residential segregation. Capital gains reinforce these geographic disparities through at least two channels. First, wealth bequests—amplified by the step-up in basis at death, which eliminates the tax on unrealized gains—allow high-gain families to transfer substantial resources to the next generation in a geographically concentrated manner. Second, real estate appreciation affects local public finance: because property taxes fund the majority of K–12 education, areas with rising home values generate more per-pupil revenue, potentially widening educational quality gaps between high-gain and low-gain regions.

The geographic concentration of capital gains may itself be partly endogenous to state tax policy. States without income taxes—including Florida, Texas, Nevada, Wyoming, Tennessee, and South Dakota—attract high-wealth individuals seeking to minimize their tax burden on realized gains. **?** document that state taxes significantly affect the location decisions of "star" scientists and inventors, with a 1 percent increase in the top tax rate reducing the stock of star scientists by 1.6 percent. **?** find that millionaire tax migration, while modest in aggregate, is concentrated among the highest-income households and responsive to cross-state tax differentials. This sorting behavior reinforces the geographic concentration of capital gains: Florida's appearance in the top five states for overall gains is likely driven in part by the in-migration of high-wealth retirees attracted by the absence of state income taxes, rather than solely by local asset appreciation.

Finally, the geographic concentration of capital gains creates heterogeneous local wealth effects that amplify regional business cycle asymmetries. **?** show that declines in housing wealth during the Great Recession had large effects on household consumption, with a marginal propensity to consume out of housing wealth of approximately 5–7 cents per dollar. Because capital gains—and the underlying asset holdings—are geographically concentrated, asset price booms disproportionately stimulate consumption and employment in already-prosperous areas, while busts devastate less-diversified regions. The experience of Nevada and Arizona during the 2008 housing crash illustrates this vulnerability: states with outsized exposure to real estate gains experienced the most severe contractions when prices reversed. The time-series volatility of capital gains at the state level thus represents a significant source of asymmetric regional risk, with implications for both macroeconomic stabilization policy and the design of fiscal federalism.